

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

NAME: SOLUTIONS CLASS: IW / HM MARK:

Achievement Standards being tested

Student compares processes of rock formation, including the time scales involved.

Mark	ND	NW	C	HC	O
Mark Range	0-9	10 - 17	18 - 25	26 - 31	32 - 40

Each of the multiple choice questions is worth one mark.

QUESTION	ANSWER
1	A
2	B
3	D
4	A
5	C
6	C
7	C
8	C
9	D
10	D
11	C
12	D
13	D
14	C
15	C
16	A
17	D
18	A
19	D
20	A

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

The following table can be used to answer questions 1 - 5.

Mineral	Streak	Hardness	Lustre	Colour
Diamond	No streak	10	Glassy	Colourless
Quartz	White	7	Glassy	White
Calcite	White	3	Glassy	White
Talc	White	1	Glassy	White

- 1 Which is the hardest mineral?
 A diamond
 B quartz
 C calcite
 D talc
- 2 Which of the following metals is also a mineral?
 A galena
 B gold
 C clay
 D aluminium
- 3 Another name for the 'shininess' of a mineral is:
 A streak
 B transparency
 C crystal
 D lustre
- 4 Why does diamond produce no streak but the other minerals do?
 A The diamond is too hard, it does not leave a mark on the porcelain tile.
 B The diamond is too soft, it crumbles instead of leaving a mark.
 C The diamond is colourless.
 D It does leave a streak, but only on other minerals.
- 5 If you were given just the three white minerals what test would you do to identify which is which?
 A Perform the streak test on a black porcelain tile.
 B Add acid to each mineral and see what happens.
 C Rub the minerals against one another to see which one is able to scratch the others.
 D Compare the shape of the crystals.
- 6 When minerals form a definite shape and their surfaces have faces we say the mineral has formed:
 A an element
 B a compound
 C a crystal
 D a cleavage
- 7 Which of the following statements concerning minerals is NOT true?
 A naturally occurring
 B solid
 C organic
 D formed of atoms
 E definite chemical structure

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

- 8 Which of the following statements concerning rocks is NOT true?
A naturally occurring
B solid
C definite chemical structure
D composed of minerals
- 9 In general, igneous rocks form:
A at relatively high temperatures
B by crystallisation of a solid from a liquid
C at high and low pressures
D under all the conditions listed above
- 10 Aside from mineral sediment, what else can form sedimentary rocks?
A shells and hard parts of sea organisms
B dead plant material
C chemicals left from the evaporation of water
D all of the above
- 11 Rocks that have been formed from other rocks due to intense heat and pressure are called:
A sedimentary
B igneous
C metamorphic
D mineralised
- 12 The rock cycle shows that:
A sedimentary rocks are formed by the compaction and cementation of sediments
B metamorphic rocks can be formed by the action of heat and pressure
C igneous rocks are formed by cooling magma
D over long periods of time rocks can be changed from one type to another by a variety of different processes
- 13 The age of the Earth is currently thought to be:
A about 6 thousand years old
B about 6 billion years old
C about 4.5 million years old
D about 4500 million years old
- 14 The geological time scale is useful because it:
A tells us the age of the Earth
B shows how different life forms have lived on Earth at different times
C divides the history of the Earth into different time periods based on major events such as ice ages, volcanic activity or mass extinctions of species
D enables the ages of fossils to be determined
- 15 The primary difference between magma and lava is that:
A magma becomes lava anytime it is in a cooling condition
B they are produced from different parent materials
C one is below the ground; the other is above the ground
D one contains more iron and oxygen than the other
- 16 Which of the following will not make a fossil?
A decomposed organic material
B plant impressions
C animal footprints
D animal bones
- 17 Coal is formed when:
A muds rich in decomposing plant and animal material settle at the bottom of oceans
B oil is compressed into a solid
C plant remains are exposed to covered with water
D plant remains are buried by sediments in swamps

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

- 18 _____ is a well known example of a rock formed under the Earth when magma cools.
A Granite
B Gneiss
C Basalt
D Pumice
- 19 The sticking together of sediment particles by dissolved chemicals is called:
A compaction
B sedimentation
C deposition
D cementation
- 20 Most fossils are found in rocks that were:
A formed at the bottom of seas or lakes
B formed by the cooling of lava from volcanoes
C changed by heat and pressure
D in sand at the seashore and in deserts

Short Answer Questions - Each questions shows the number of marks allocated. The amount of space given for each answer should tell you how much you need to write for an answer.

- 21 What does the word 'extinct mean'? (1 mark)
• No longer living on Earth. (1)
- 22 What is an ore? In which ore would you find iron? (1 mark)
ORE - mineral with a large amount of useful metal in it. (1)
IRON - found in haematite (iron ore). (1)
- 23 What is the difference between intrusive and extrusive igneous rocks? (2 marks)
INTRUSIVE - large crystals; formed below the surface. (1)
EXTRUSIVE - small crystals; formed on the surface. (1)
- 24 There are a few different types of basalt dependent on how fast it forms outside the volcano. What determines which type is going to be formed? (2 marks)
• How fast it cools. (1)
• The type of eruption. (1)

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

The next four questions (25–26) refer to the following information:

Mineral	Chemical formula	Hardness	Uses
Quartz	SiO_2	7	Melted and made into glass and computer chips
Talc	$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$	1	Paint, ceramics, paper and talcum powder
Gold	Au	2.5–3	Dentistry, jewellery, computers and coins
Galena	PbS	2.5	Lead is obtained from this mineral. Lead is used in car batteries and also lead sheet and pipes
Silver	Ag	2.5–3	Jewellery, cutlery, coins, electronics and photography
Zirconia	ZrO_2	8.5	Imitation diamonds, used in jewellery

- 25 I have found a beautiful lump of quartz. I put it into my box of precious things where my gold and silver jewellery is stored. Why is this NOT a good idea? (2 marks)

Quartz is much harder than gold and silver. (1)

Will scratch them. (1)

- 26 When gold is used to make jewellery it is usually mixed into an alloy with other metals. Why is this necessary? (2 marks)

Gold is soft and scratches easily. (1)

An alloy is harder and resists scratching (1).

**YEAR 8 SCIENCE
EARTH SCIENCES
TEST**

The next two questions (27–28) refer to the following information:

You are a forensic science investigator. You have been called to assist with solving a diamond robbery case. The prime suspect is K.G. Crook. He has been apprehended near to the scene of the crime and he has many beautiful shiny stones in his possession. K.G. claims that these stones are cheap imitation diamonds, called cubic zirconia that he has purchased from a friend. You suspect that these stones are the stolen diamonds.

- 27 What simple test could you carry out to prove that the stones in K.G.'s possession are in fact diamonds? (2 marks)

Scratch test. (1)

Zirconia is softer than diamond and scratches easily.

- 28 If these shiny stones are imitation diamonds what result would you expect to get from the test? (2 marks)

Scratching with diamond will damage zirconia. (1)

Using topaz + corundum (from table before) shows it to be 8.5 on Moh's scale. (1)

- 29 What is difference between weathering and erosion? Use examples of where each occurs in a natural setting? (4 marks)

WEATHERING: occurs slowly over time; can be due to (1)

chemicals (acids) or moving particles (due to water and wind), e.g. Rocks go smooth in rivers. (1)

EROSION: shifting of a large area of soil by wind or water. (1)

e.g. Floods washing away top soil. (1)